

Edimah SYNESIUS SONGO

Applied Mathematician

📍 Paris @ edimah.synesiussongo@proton.me

🔗 edimah.github.io 🐙 github.com/Edimah in linkedin.com/in/edimah-synesius

Bayesian inference, calibration, uncertainty quantification, sensitivity analysis and hypothesis testing, applied to model evaluations. Interest: measurements from inside the model that hold up under outside scrutiny.

Methods and tools. Python · R · SQL · C++ · Git, $\text{\LaTeX}$ · Bayesian inference & MCMC · calibration · uncertainty quantification · sensitivity analysis · permutation/bootstrap inference · multiple-comparison correction

🔬 SELECTED WORK

DETECTING AGENTIC COLLUSION IN PUBLIC LOGS WITH STATISTICAL METHODS

APART RESEARCH × CESIA SPRINT, SEPT 2026

Track 2. 14,591 reconstructed agent revisions and 32,656 live edits over four wikis, 2003–2026. A ladder of four null models, up to a difference-in-differences against an untouched sibling wiki with a placebo-in-time test: the incident weeks exceed all 4,872 reference weeks, the effect is 3.2 log points, the change point falls on 16 June. A weekly check built from the same features raises no false alarm over the 2,236 wiki-weeks of the preceding decade.

🐙 github.com/Edimah/agent-collusion-detector-in-public-logs

A CALIBRATED TWO-LEVEL DETECTOR FOR SECRET LOYALTIES

APART RESEARCH × FORMATION RESEARCH SPRINT, JULY 2026

With A. Randrianandrasana. My contributions: estimator, permutation test, calibration bench, difference-in-differences, control selection, activation probe. Calibration-matched controls cut false positives from 14/16 to 2/16 on a clean model, where every positive is false by construction. The instrument specification was written before any result existed and kept as a pre-registration, its result fields never back-filled. Reviewer: “This is rigorous work, and the calibration is a key contribution here.”

🔗 apartresearch.com/project/a-calibrated-twolevel-detector-for-secret-loyalties-g75x

🐙 github.com/Edimah/secret-loyalties

MODELLING MNAR DATA WITH POPULATION DATA #1 : WHY IS THIS CELL BLANK?

BLOG POST, JULY 2026

Suppression in published cancer-registry tables reads the count it hides, which makes the missingness not-at-random. Across tenths of expected case count the share of withheld cells runs from 93% to 0%; across tenths of population, from 13% to 1%.

🔗 edimah.github.io/blog/2026/why-is-this-cell-blank

📁 EXPERIENCE

Present
February 2026

Interim Data Lead, CRCDC GUADELOUPE, Regional cancer screening

- Rebuilding the regional cancer-screening data chain to deliver reliable data to stakeholders.
- Delivered a data-chain audit and methodology documentation under regulatory constraints.

R SQL eSIS DataQuery Data Management Health Data Regulation

Present
October 2025

Independent Consultant, APPLIED STATISTICS, AI Implementation

- Feasibility analysis of AI tools in regulated health care (pharmacy) to facilitate professional workflows.

Data Management Communication Data Visualisation

September 2024
November 2023

Doctoral Researcher, IFP ENERGIES NOUVELLES / ONERA / ENSAI, Île-de-France

- Multidisciplinary sensitivity analysis for complex physics systems.
- Left the doctorate before completion.

R Python Sensitivity Analysis Uncertainty Quantification

September 2023
April 2023

Research Intern, EDF R&D, Chatou

- *M2 Thesis: Advanced methods for model calibration with uncertain inputs.*
- Designed adaptive, MCMC-based parameter re-adjustment strategies to handle uncertainty.

Python OpenTURNS Bayesian Inference MCMC Algorithms

🎓 EDUCATION

2023 **Master 2 — MASH** (Mathematics, Machine Learning, Sciences, Humanities), **Université PSL** (Dauphine & ENS), with Merit.

2022 **Master 1 — Applied Mathematics**, Statistics track, **Paris Dauphine-PSL**.

2021 **Bachelor's — MEFA** (Mathematics, Economics, Finance, Actuarial Science), **Paris Dauphine-PSL**.

❤️ HOBBIES

String instruments · Boulderling · Muay Thai